EXECUTIVE REPORT · DATA ANALYTICS PORTFOLIO

Global Coffee Market Analysis

Production · Exports · Imports · Domestic Consumption

Period Covered: 1990 – 2019

90 Countries | 30-Year Span | 4 Core Metrics | Tableau Dashboard

TOTAL PRODUCTION	TOTAL EXPORTS	TOTAL IMPORTS	LEADING PRODUCER	PURE IMPORTERS
225.7B	**163.3B**	**177.4B**	**Brazil**	**35**
60-Bag Units	60-Bag Units	60-Bag Units	Consistent 1990-2019	Countries – Zero Production

Built with:

- Python (Pandas · NumPy · Matplotlib)
- Jupyter Lab (EDA & Data Prep)
- Tableau (Interactive Dashboard)

"Coffee is the world's most traded tropical commodity — understanding its flows is understanding the pulse of global agricultural markets."

Data Analysis & Dashboard Design

Data Analytics Portfolio Project · Ezequiel Gonzalez

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Project Context

This report documents a full end-to-end data analytics project: raw data was ingested and transformed in **Python** / **Jupyter Lab**, then visualised in an interactive **Tableau dashboard**. This document translates those technical outputs into a consultant-grade executive narrative aimed at business and portfolio audiences.

Key Technical Contributions

Built an end-to-end ETL workflow in Python

Standardized country identifiers across 4 datasets

Resolved malformed sentinel values

Designed an interactive Tableau dashboard

Developed country-level exploratory navigation

Created reusable segmentation logic

OVERVIEW OF FINDINGS

1 Executive Summary

This report presents a comprehensive analysis of global coffee market dynamics across 30 years (1990–2019), encompassing **90 countries** and four core trade metrics: production, exports, imports, and domestic consumption. The study was conducted using a structured analytics pipeline — exploratory data analysis (EDA) in Python followed by interactive visual storytelling in Tableau.

The global coffee market has undergone a fundamental structural shift in the period under study. Total production grew from approximately **5.6B units in 1990 to nearly 9.9B in 2019** — a 77% increase driven primarily by three countries. Meanwhile, import volumes expanded steadily, reflecting an ever-growing gap between where coffee is grown and where it is consumed.

Five Executive-Level Findings

1 Brazil's Unrivalled Dominance

Brazil has consistently accounted for roughly 30–39% of global coffee production, leading exports (40.3B total) and domestic consumption (27.8B) simultaneously — a unique position in any global commodity.

2 Vietnam's Structural Transformation

Vietnam expanded production from 78.6M units in 1990 to over 1.8B by 2019 — a 23x increase — becoming the world's second-largest producer and reshaping the global supply landscape.

3 A Bifurcated World Market

The dataset reveals a clean divide: 55 countries produce coffee; 35 countries — primarily developed economies — import without any domestic production. This structural asymmetry defines global coffee dependency.

4 The United States: Largest Importer

With 42.5B units imported over 30 years and zero domestic production, the USA stands as the world's largest pure importer, followed by Germany (31.5B) and Italy (13.3B).

5 Supply vs Demand Divergence

While production is geographically concentrated in Latin America, Africa, and South-East Asia, consumption is increasingly globalised. Import volumes grew steadily from 4.4B (1990) to over 8.1B (2019), reinforcing structural dependency in consumer markets.

WHY THIS ANALYSIS MATTERS

2 Business Problem

Coffee is one of the world's most traded agricultural commodities, second only to crude oil in dollar value. Understanding the structural dynamics of who produces, who exports, and who depends entirely on imports is strategically relevant for commodity traders, agricultural policy-makers, sustainability analysts, and investment teams operating in agri-food markets.

Core Business Questions

Q1. Which countries dominate global coffee production?
Identifying concentration risk and supply-side power dynamics.

Q2. Which countries are pure importers with zero production?
Revealing structural demand dependency and geopolitical supply vulnerabilities.

Q3. How has the production landscape shifted over three decades?
Understanding macro-level structural changes in the global coffee supply chain.

Q4. What is the relationship between production volume and export volume?
Quantifying export orientation and domestic consumption-retention ratios.

Q5. What regional patterns exist in coffee trade flows?
Enabling geographic segmentation of market participants.

Strategic relevance for a data analyst: This project demonstrates the ability to move from raw multi-source data to actionable business intelligence — translating agricultural trade statistics into a decision-support tool for commodity strategy, market entry analysis, and supply chain risk assessment.

DATASET STRUCTURE AND SCOPE

3 Data Overview

The analysis is built on the **coffee_master** dataset — a consolidated master table produced during the EDA phase by merging four International Coffee Organisation (ICO)-sourced CSV files covering production, domestic consumption, exports, and imports. The merge was performed in Python using country and year as joint keys.

Dataset Specifications

Attribute	Detail
Total Rows	2,700

Countries Covered	90
Year Range	1990 – 2019 (30 years)
Core Metrics	Production, Consumption, Exports, Imports
Unit of Measure	60-Kg Bags (as per ICO standard)
Source Files	4 raw CSVs (production, consumption, exports, imports)
Null Rate – Imports	61.1% (35 pure-importing countries have no production/export data)
Null Rate – Production	38.9% (production data absent for pure-importer nations)

Data Quality Notes

- Import records contained a sentinel value of −**2,147,483,648** (INT32 min), representing missing data. These were replaced with NaN during the EDA phase.
- Production and consumption columns use a crop-year convention ('1990/91') which was normalised to a single integer year (e.g., 1990) during wrangling.
- Country name inconsistencies (trailing whitespace, alternate spellings) were detected via set-difference checks between source files and corrected via string-stripping.
- 61.1% of Imports rows are null — a structurally valid pattern, since producing countries generally do not import coffee. These nulls are **meaningful absences**, not data errors, and were preserved intentionally in the dashboard logic.
- The final coffee_master CSV was exported as the clean analytical base for all Tableau calculations.

Analytical Decision Making

Missing values were intentionally preserved where they represented genuine absence of activity rather than data quality issues.

This prevented artificial inflation of trade metrics and ensured analytical transparency.

Data Challenges & Solutions

Challenge	Investigation	Solution	Impact
Sentinel values in exports	Export records contained -2147483648	Converted to NaN	Prevented distortion of trade metrics
Inconsistent country names	Country names differed across source files	Harmonized identifiers	Enabled accurate joins
Crop-year formatting	Production used 1990/91 format	Extracted starting year	Unified time dimension
Missing export observations	Brazil 2014–2015 exports absent	Preserved as NULL	Maintained analytical integrity

EDA PIPELINE + TABLEAU DESIGN LOGIC

4 Methodology

The project follows a two-stage analytics pipeline: a **Python-based exploration and preparation phase** (implemented in Jupyter Lab), followed by a **Tableau visualisation phase**. This separation of concerns ensures analytical rigour in the data layer before visual design decisions are made.

Stage 1 — Exploratory Data Analysis (Python / Jupyter Lab)

- **Data Ingestion:** Four raw CSV files (production, consumption, exports, imports) were loaded independently into Pandas DataFrames.
- **Reshape (Wide → Long):** Year-column-based wide tables were melted into long format using pd.melt(), with year columns identified by slash notation ('1990/91') for production/consumption and numeric strings for trade data.
- **Year Normalisation:** Crop-year strings ('1990/91') were split to extract the start year and cast to int64 for consistent joining.
- **Country Harmonisation:** String-stripping applied to all Country fields; set-difference operations between DataFrames identified countries exclusive to only one dataset.
- **Sentinel Value Replacement:** INT32 minimum (−2,147,483,648) replaced with NaN in the Exports column.
- **Master Merge:** Four long-format DataFrames joined sequentially on [Country, Year] using outer joins to preserve all observations.
- **Derived Metrics:** Self-Sufficiency Ratio (Production ÷ Consumption) and Export Intensity (Exports ÷ Production) computed for producing-country subset.
- **Country Segmentation:** Producer vs importer classification based on Production nullability — 55 producing countries vs 35 pure-importer nations.
- **Outlier / Infinity Handling:** np.inf values arising from division by zero (e.g., countries with zero consumption) replaced with NaN before aggregation.
- **Export:** Cleaned coffee_master.csv saved as the analytical output and input for Tableau.

Stage 2 — Tableau Dashboard Design

- **Map Encoding — Production as Primary Dimension:** Production volume was chosen as the choropleth colour driver. This decision maximises clarity: the most strategically important metric (supply) is immediately visible, while the dark black base map makes producing countries visually prominent.
- **Null-Aware Tooltip Logic:** Tooltips show all four metrics but conditionally display Imports only when a non-null value exists. This prevents misleading empty import fields for producing nations.
- **Pure Importer Identification:** Countries with null Production but non-null Import values are flagged as 'pure importers'. The USA tooltip (Imports: 42.5B, Production: —) is the canonical example.
- **Colour Palette:** Warm earth tones (cream → dark red) reinforce the coffee thematic while maintaining perceptual accuracy — darker = higher production.
- **Coffee Market Dynamics Chart:** A multi-series line chart displays annual global totals for Production, Exports, Consumption, and Imports from 1990–2019, enabling temporal trend analysis alongside the geographic map.
- **KPI Strip:** Four summary KPIs (Total Production, Total Exports, Leading Producer, Largest Consumer) appear at the top of the dashboard as at-a-glance executive metrics.

5

FIVE CRITICAL MARKET FINDINGS

Key Insights

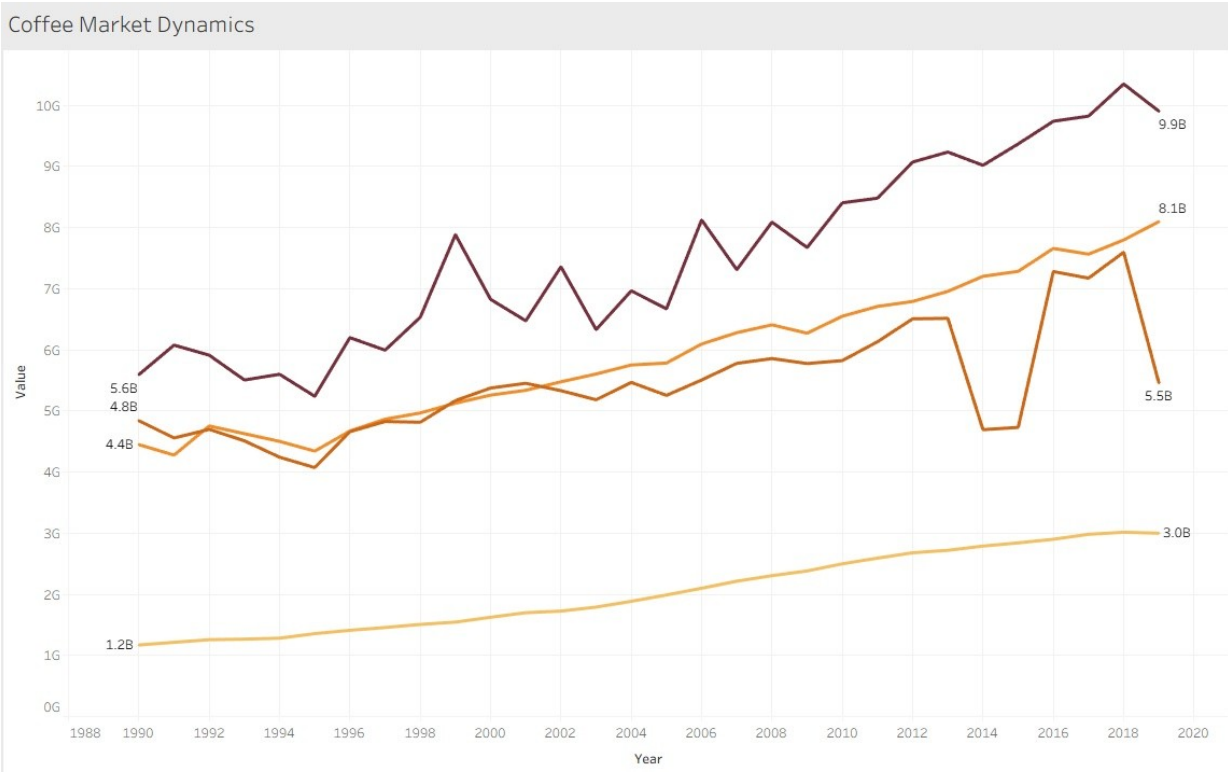

Figure 1 — Coffee Market Dynamics (1990–2019): Production (dark red) vs Exports (orange) vs Imports (amber) vs Consumption (light yellow). Source: Tableau Dashboard.

Insight 1 — Brazil's Structural Dominance Across the Entire Value Chain

With **75.1B units of cumulative production** (1990–2019), Brazil accounts for approximately **33% of global output** over the study period. Uniquely, Brazil is also the world's largest domestic consumer (27.8B units) and leading exporter (40.3B units) — simultaneously dominating supply and demand.

Insight 2 — Vietnam's Structural Ascent: A 23× Production Expansion

Vietnam entered the 1990s as a marginal producer at 78.6M units. By 2019, production reached **1.83B units** — a **23-fold increase in 30 years**. This is one of the most dramatic production scale-ups in agricultural commodity history, driven by government-backed Robusta cultivation programs. Vietnam now ranks as the world's second-largest producer, accounting for approximately 13% of global output by 2019.

Insight 3 — Production Concentration: Top 3 Countries = ~55% of Global Supply

Brazil, Vietnam, and Colombia collectively generated over half of total global coffee production in the study period. This extreme concentration creates systemic supply risk: climate shocks, policy shifts, or infrastructure disruption in any of these three nations could materially alter global coffee availability and pricing.

Rank	Country	Total Production (Units)	% of Top-10 Total
1	**Brazil**	75.1B	36.2%
2	**Viet Nam**	28.8B	13.9%
3	**Colombia**	21.6B	10.4%
4	**Indonesia**	15.4B	7.4%
5	**Ethiopia**	8.7B	4.2%
6	**India**	8.2B	4.0%
7	**Mexico**	7.9B	3.8%
8	**Guatemala**	7.0B	3.4%
9	**Honduras**	6.7B	3.2%
10	**Uganda**	5.9B	2.8%

Table 1 — Top 10 Coffee Producing Countries, Cumulative 1990–2019. Source: coffee_master dataset.

Insight 4 — Import Markets Are Almost Exclusively Developed Economies

Of the 35 countries classified as pure importers (zero domestic production), all are either European Union member states, or high-income OECD economies (USA, Japan, South Korea). This is not a coincidence — it reflects the fact that coffee cultivation requires specific tropical and subtropical conditions (the 'Coffee Belt' between 25°N and 30°S), creating a structural North-South trade dependency.

Rank	Country	Total Imports (Units)	Production
1	**United States of America**	42.5B	—
2	**Germany**	31.5B	—
3	**Italy**	13.3B	—
4	**Japan**	12.4B	—
5	**France**	12.0B	—
6	**Spain**	7.8B	—
7	**United Kingdom**	6.7B	—
8	**Belgium**	6.2B	—
9	**Netherlands**	6.0B	—
10	**Russian Federation**	5.7B	—

Table 2 — Top 10 Coffee Importing Countries (Pure Importers). Source: coffee_master dataset.

Insight 5 — Global Import Volumes Nearly Doubled Over 30 Years

Global imports grew from **4.4B units in 1990 to 8.1B in 2019** — a 84% increase that closely mirrors production growth. This parallel expansion confirms that coffee trade has not merely redistributed existing supply — it has genuinely scaled, driven by rising middle-class consumption in previously low-consuming markets (notably China, South Korea, and Eastern Europe).

Country Segmentation

During exploratory analysis, it became clear that countries played fundamentally different roles within the global coffee ecosystem. To better understand these differences, countries were grouped into three structural categories based on their production, export and import behavior.

PRODUCERS VS EXPORTERS VS PURE IMPORTERS

6 Country Segmentation

A central analytical output of this project is a three-way segmentation of all 90 countries in the dataset. This classification was defined during the EDA phase and applied as the structural logic of the Tableau dashboard.

Segment	Definition	Count	Key Examples
Production Leaders	Countries with Production > 0 and Exports > 0; supply-side dominance in global trade.	15	Brazil, Viet Nam, Colombia, Indonesia, Ethiopia
Producing & Consuming	Countries producing coffee but retaining most for domestic consumption; lower export intensity.	40	Mexico, Peru, India, Honduras, Uganda
Pure Importers	Countries with null Production and null Exports; entirely dependent on the global trade network.	35	USA, Germany, Italy, Japan, France, UK

Table 3 — Country Segmentation Framework. Derived from dashboard logic and EDA outputs.

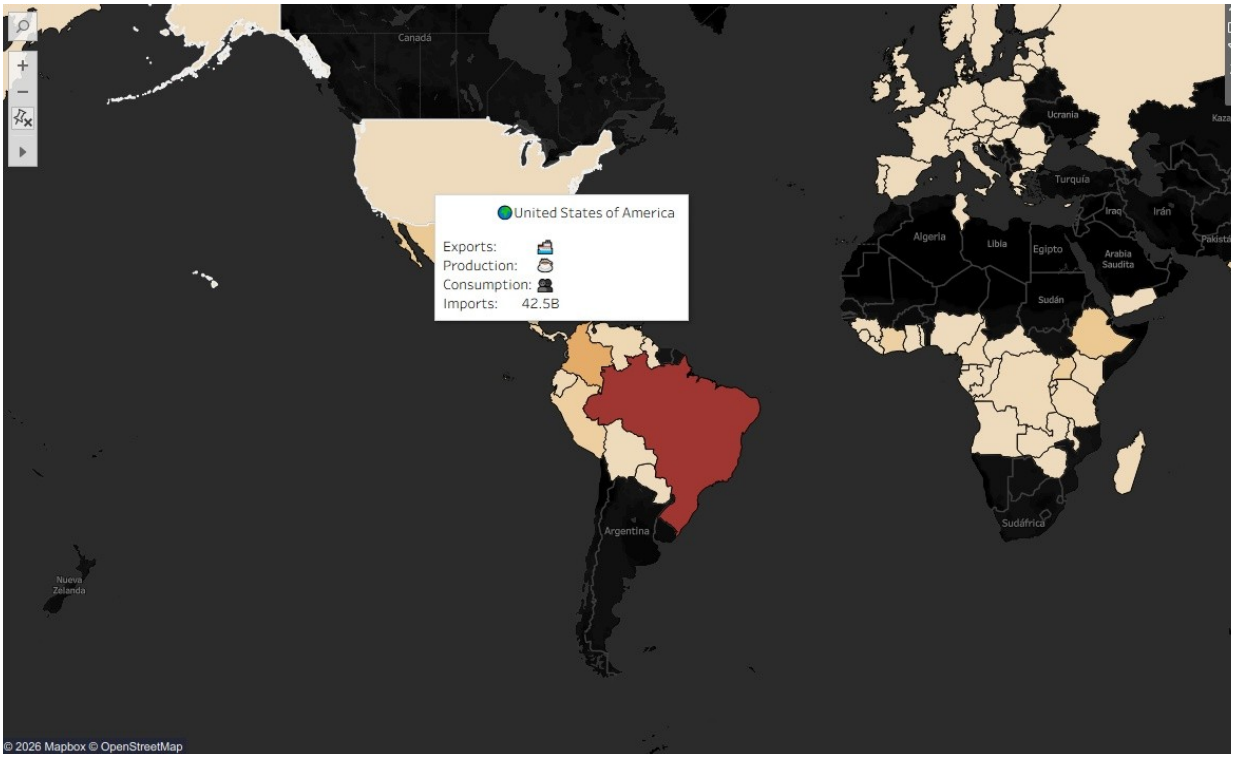

Figure 2 — Tableau Map: United States tooltip showing 42.5B units imported with null Production and Exports — canonical example of a pure importer country.

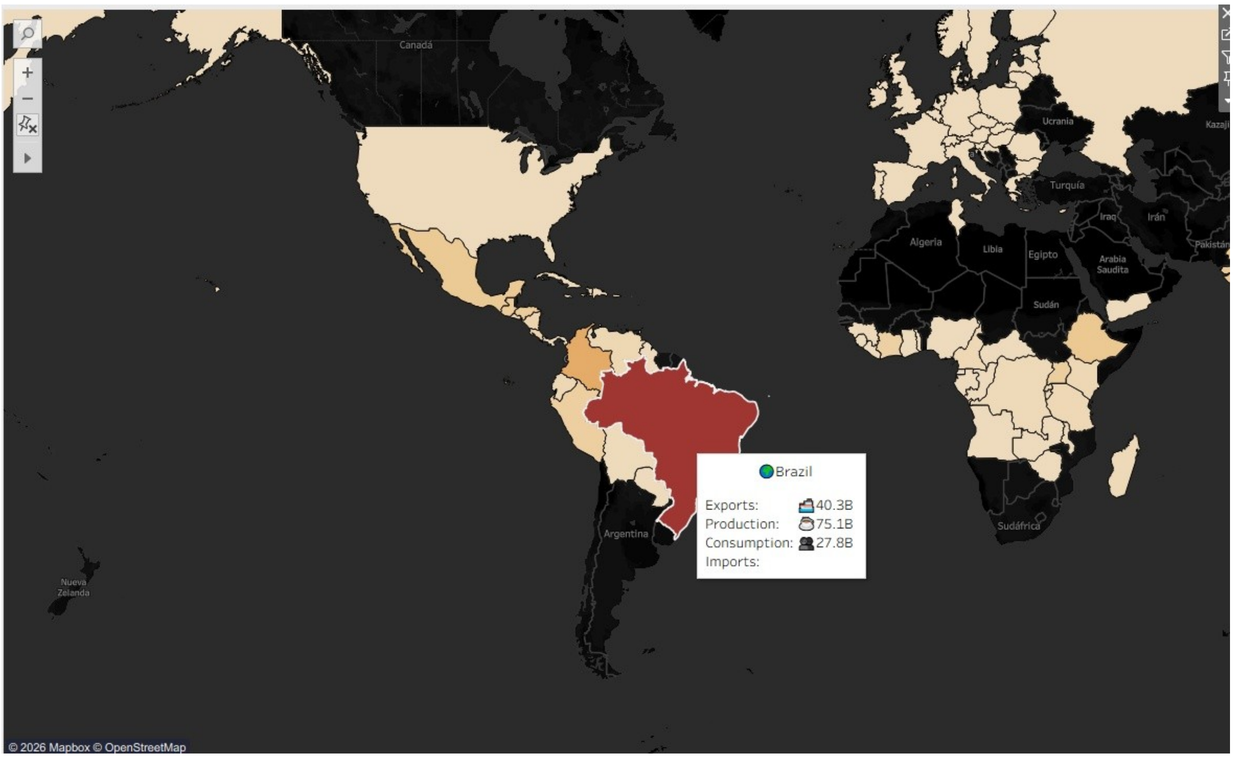

Figure 3 — Tableau Map: Brazil tooltip — Production 75.1B, Exports 40.3B, Consumption 27.8B. Imports field intentionally absent (null), reflecting producer-country logic in the dashboard.

Export Intensity Analysis

Among producing countries, a key secondary metric is **Export Intensity** (Exports ÷ Production). Countries like Equatorial Guinea, Togo, Burundi, and Cameroon show Export Intensity above 1.0 — meaning they export more than their recorded production, suggesting re-export activity or reporting discrepancies in the ICO data. Vietnam and Uganda, by contrast, show high but sub-1.0 intensities, consistent with predominantly export-oriented production with modest domestic retention.

INTERACTIVE TABLEAU DESIGN EXPLAINED

7 Dashboard Walkthrough

The Tableau dashboard was designed as a single-view, multi-component executive interface. It combines a global choropleth map with a multi-series line chart and a KPI strip — all driven by the coffee_master dataset.

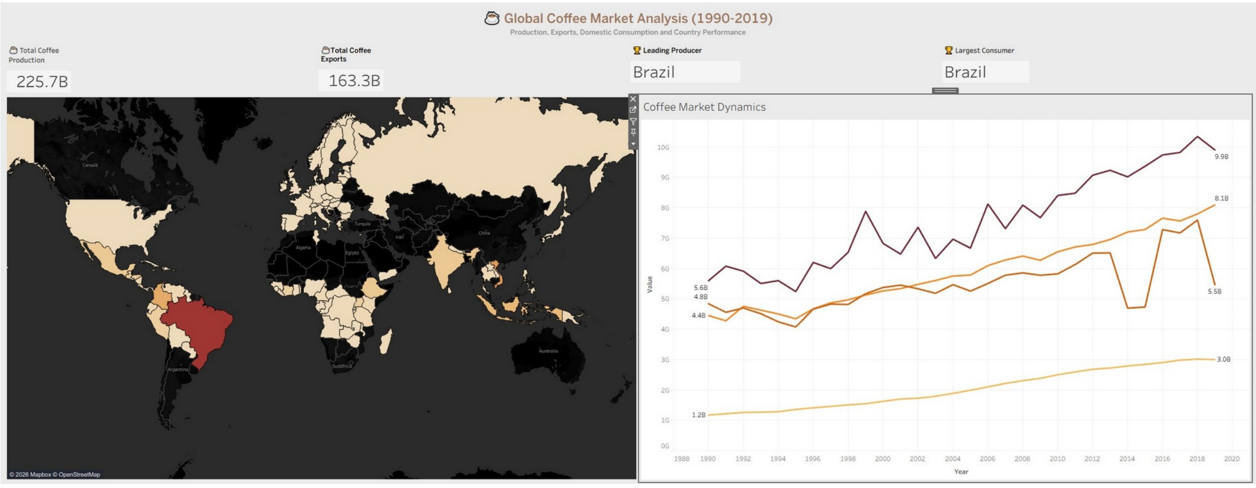

Figure 4 — Full Dashboard View: KPI strip (top), interactive world map (left), and Coffee Market Dynamics trend chart (right). Source: Tableau dashboard built on coffee_master dataset.

Component 1 — KPI Summary Strip

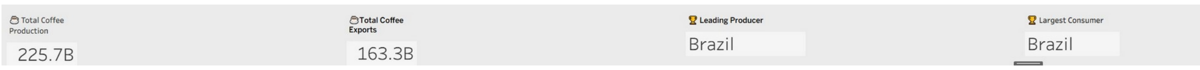

Figure 5 — KPI Strip: Total Production (225.7B), Total Exports (163.3B), Leading Producer (Brazil), Largest Consumer (Brazil). Source: Tableau.

Four headline KPIs appear at the top of the dashboard, providing an immediate executive summary before the user engages with the interactive map. The KPIs are computed from the full 1990–2019 dataset and update in real time with any applied filters.

Component 2 — Interactive World Map

The choropleth map colour-encodes countries by cumulative Production using a warm gradient from cream (low/null) to dark red (high). Countries with zero production appear in a very light neutral, immediately distinguishable from the producing nations.

- **Tooltip Design:** Clicking or hovering on any country reveals a contextual tooltip with the country flag emoji, name, and all four metrics (Production, Exports, Consumption, Imports).
- **Conditional Imports Display:** The Imports field only appears in the tooltip when a non-null value exists — eliminating blank rows for producing countries where imports data is structurally absent.
- **Dark Map Base:** A dark Mapbox basemap (black background) maximises contrast for the warm coffee palette overlay, creating a high-impact visual presentation.

Component 3 — Coffee Market Dynamics Chart

The right panel shows a multi-series line chart of annual global totals for all four metrics (1990–2019). Key observations visible from the chart:

- Production (dark red line) peaks near 10.3B in 2018 before a slight retreat.
- Exports (orange) trend steadily upward, demonstrating that production growth has been matched by expanding export capacity.
- Imports (amber) track almost perfectly with Exports — confirming market equilibrium at the global level, as expected in a commodity trade dataset.
- Domestic Consumption (light yellow) grows slowly and steadily, rising from 1.2B (1990) to 3.0B (2019) — a 150% increase over 30 years.
- The 2014 anomaly in the Exports series (visible dip) is a notable data irregularity flagged during EDA — likely a reporting gap in the ICO export dataset for that crop year.

DECISION-MAKING IMPLICATIONS

8 Strategic Conclusions

The following conclusions are derived directly from the analytical outputs and are framed as decision-relevant statements for business stakeholders — not technical observations.

1. Supply Chain Risk is Highly Concentrated

With three countries (Brazil, Vietnam, Colombia) supplying over 55% of global coffee, any supply-side shock — whether climate, political, or logistical — creates immediate global market implications. Commodity risk models must treat Brazil as a systemic exposure, not merely a major supplier.

2. Pure Importers Face Structural Dependency

35 countries — all developed economies — have zero production capacity and are fully exposed to global coffee price volatility and supply chain disruptions. This creates a permanent structural bargaining asymmetry between producing and consuming nations.

3. Vietnam Represents a Case Study in Deliberate Market Entry

Vietnam's 30-year transformation from 78.6M to 1.8B units demonstrates that deliberate state investment in agricultural infrastructure can fundamentally reshape a global commodity landscape within a single generation.

4. Import Growth Is Sustainable — Consumption Growth Is the Limiting Factor

Import volumes have grown in lockstep with production, but domestic consumption growth has been slower, suggesting that the structural opportunity for emerging consumer markets (e.g., China, Eastern Europe) is not yet fully realised in the 1990–2019 window.

5. This Analysis Framework Is Scalable to Any Agricultural Commodity

The production-vs-consumption segmentation logic, the export intensity metric, and the pure-importer identification method are directly transferable to any agricultural commodity dataset (cocoa, tea, rubber). The methodology is the reusable asset.

> ***"Brazil is not only the world's largest coffee producer and exporter, but also its largest domestic consumer — making it a uniquely dominant actor across the entire coffee value chain."***

— Derived from EDA notebook Key Findings, coffee_master dataset

Project Summary

This project demonstrates a complete data analytics workflow: raw data ingestion → data cleaning → EDA in Python → derived metrics (Self-Sufficiency, Export Intensity) → country segmentation → Tableau dashboard design → executive storytelling. All findings are grounded in the coffee_master dataset; no data was imputed or fabricated.

Tools & Technologies

Tool / Library	Role in Project
Python (Pandas, NumPy)	Data ingestion, reshaping, cleaning, EDA, derived metrics
Jupyter Lab	Interactive notebook environment for the EDA phase
Matplotlib	Exploratory visualisation during EDA (bar charts, line plots)
Tableau Desktop	Interactive dashboard and choropleth map visualisation
coffee_master.csv	Processed master dataset — input for Tableau
ICO (International Coffee Organisation)	Original data source (production, trade, consumption)

This project demonstrates:

- Data Cleaning
- Data Wrangling
- Exploratory Data Analysis
- Data Quality Investigation
- Feature Engineering
- Tableau Dashboard Development
- Interactive Visual Analytics
- Business Storytelling

END OF REPORT · Global Coffee Market Analysis (1990–2019) · Portfolio Project